

POWER SUPPLIES · +3V3 from J1 pin 22 → 2.8 V (analog/pixel) · 1.8 V (IO) · 1.2 V (core)

ENABLE & SEQUENCING · one wire from J1 pin 17, RC-delayed

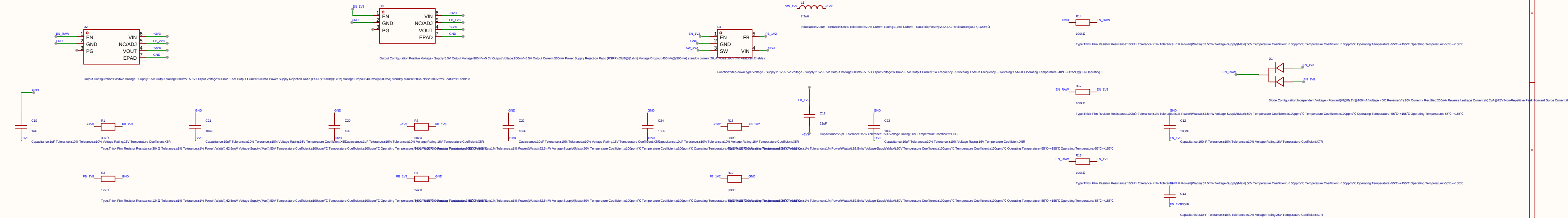
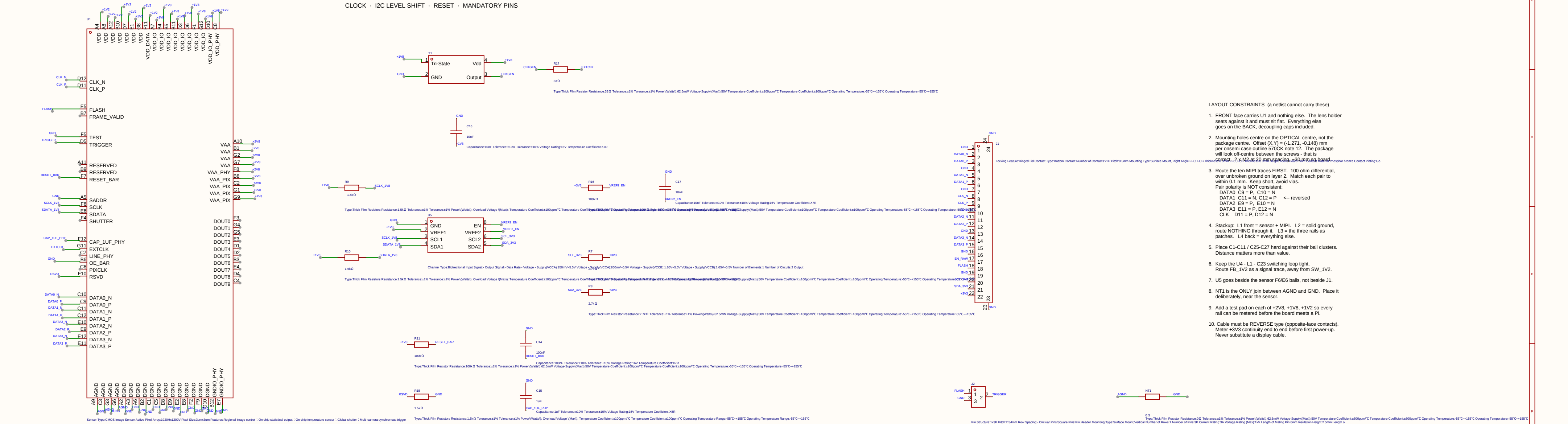


IMAGE SENSOR · U1 AR0234CS, 83-ball CSP, 1920 x 1200 global shutter, mono

CONNECTORS · J1 22-way FFC to the Pi · J2 flash/trigger header · NT1 AGND-GND tie



LAYOUT CONSTRAINTS (a netlist cannot carry these)

- FRONT face carries U1 and nothing else. The lens holder seats against it and must sit flat. Everything else goes on the BACK, decoupling caps included.
- Mounting holes centre on the OPTICAL centre, not the package centre. Offset (X,Y) = (-1.271, -0.148) mm per onsemi case outline 570CK hole 12. The package will look off-centre between the screws - that is correct - 2 x M2 at 20 mm spacing, not 30 mm as boards.
- Route the ten MIPI traces FIRST. 100 ohm differential, over unbroken ground on layer 2. Match each pair to within 0.1 mm. Keep short, avoid vias. Pair polarity is NOT consistent.
DATA0 C9 = P, C10 = N
DATA1 C11 = N, C12 = P <-- reversed
DATA2 E9 = P, E10 = N
DATA3 E11 = P, E12 = N
CLK D11 = P, D12 = N
- Stackup: L1 front = sensor + MIPI. L2 = solid ground, route NOTHING through it. L3 = the three rails as patches. L4 back = everything else.
- Place C1-C11 / C25-C27 hard against their ball clusters. Distance matters more than value.
- Keep the U4 - L1 - C23 switching loop tight. Route FB_IV2 as a signal trace, away from SW_IV2.
- U5 goes beside the sensor F0/E6 balls, not beside J1.
- NT1 is the ONLY join between AGND and GND. Place it deliberately, near the sensor.
- A test pad on each of +2V8, +1V8, +1V2 so every rail can be metered before the board meets a Pi.
- Cable must be REVERSE type (opposite-face contacts). Meter +3V3 continuity end to end before first power-up. Never substitute a display cable.

DECOUPLING · one 1 uF per supply-ball cluster · ALL of these mount on the BACK of the board, via straight through to the ball they serve

